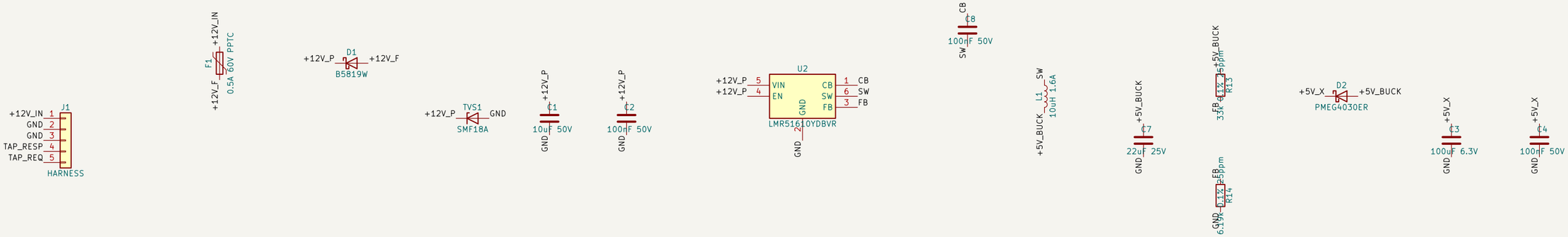
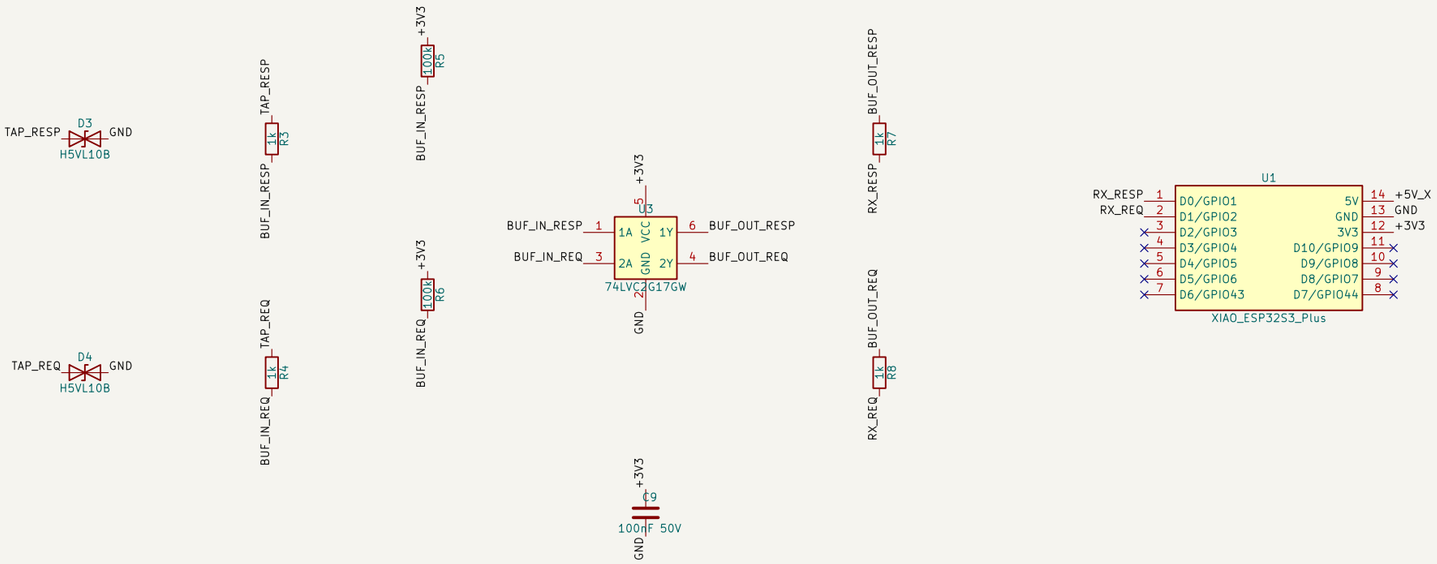
POWER: harness 12 V -> F1 PPTC -> D1 reverse-polarity -> TVS1/C1/C2 -> U2 buck 5.065 V -> D2 backfeed block -> C3/C4 -> XIAO 5V



TAPS (PASSIVE ONLY): harness 10k (off-board) -> D3/D4 ESD -> R3/R4 1k -> U3 74LVC2G17 one-way Schmitt buffer -> R7/R8 1k -> XIAO RX.

No part lets a XIAO pin drive a tap line: U3 only conducts A -> Y. R5/R6 100k hold the buffer inputs high when the harness is idle.



12V OFF FOR USB: unplug the harness inline connector or the dongle before plugging a plain USB cable into the XIAO (XIAO 5V is fed through D2), or use a VBUS-blocking USB isolator.

J1 wire pads: 1 +12V, 2 GND (power), 3 GND (signal), 4 TAP pad 9 (inverter replies), 5 TAP pad 10 (dongle requests).

Passive Modbus tap carrier for XIAO ESP32S3 Plus (generated by scripts/gen_sch.py)

SolArk_Live_Data

Sheet: /

File: carrier.kicad_sch

Title: SolArk XIAO tap carrier

Size: A3	Date: 2026-10-13	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 1/1